

Global Climate Change — Fact Sheet

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2024 was confirmed by the World Meteorological Organization (WMO) as the warmest year on record — global average surface temperature reached about 1.55°C (±0.13°C) above the 1850-1900 pre-industrial baseline. Atmospheric CO₂ hit a record annual average of 422.8 ppm in 2024 and 427.5 ppm in 2025, roughly 50% above pre-industrial levels of ~280 ppm.

Key Greenhouse Gases (share of 2024 global GHG emissions, EDGAR)

Gas	Share	Primary Sources	Since 1990
CO ₂	74.5%	Fossil fuels, deforestation, cement	+74.9%
Methane (CH ₄)	17.9%	Livestock, landfills, oil & gas	+30%
Nitrous oxide (N ₂ O)	4.8%	Fertilizers, industrial processes	+34%
Fluorinated gases	2.8%	Refrigeration, air conditioning	+310%

Top Emitters (2024, EDGAR / Carbon Brief)

Country/Bloc	Share of GHG	YoY Change	Per-capita (tCO ₂ e)
China	~30%	+0.2% to +0.8%	~10
United States	~12-14%	-0.1% to -0.6%	~17.5
India	~7-8%	+3.9% to +4.6%	~2.5
EU (27)	~6-7%	-1.8% to -3.8%	~5.6
Russia	~4.8%	+2.4%	~14.1

The G20 accounts for ~83% of global CO₂ emissions; top 8 emitters ~61-68% of the global total.

Observed Impacts

- Global mean sea level rising ~3.3-4.5 mm/year since 1993 (accelerating)
- Arctic sea ice September minimum down to ~3.9-4.5 million km², vs. ~7.5-8M in the early 1980s
- Oceans have absorbed over 90% of excess heat and ~25% of human CO₂ emissions
- More frequent and intense heatwaves, droughts, and heavy rainfall events

Policy Timeline — Through COP30

- **1992 — UNFCCC**: founding treaty for international climate cooperation.
- **1997 — Kyoto Protocol**: first binding emissions targets, developed nations.
- **2015 — Paris Agreement (COP21)**: limit warming to well below 2°C, pursue 1.5°C.
- **2023 — COP28, Dubai**: first Global Stocktake; pledge to triple renewables by 2030.
- **2024 — COP29, Baku**: New Collective Quantified Goal on climate finance.
- **2025 — COP30, Belem**: "Belem Package" adopted by 195 Parties — adaptation finance tripling pledge by 2035, Just Transition Mechanism, ~\$6.6bn forest fund; binding fossil-fuel roadmap not adopted.

The Energy Transition (2024)

Metric	2024 Value
Renewable capacity added	582-585 GW (record, +15.1% YoY)
Total installed renewable capacity	~4,448 GW (~46% of global power capacity)
Solar PV / Wind added	~452-453 GW / ~113-114 GW
China's share of new capacity	~64%
2030 target (COP28 pledge)	11.2 TW (needs ~16.6%/yr growth)

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